

Anomaly detection, beyond tables

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This afternoon

- **ISD-1020** · Monday 28 September 2026 · PUIO B210
- 13:30–17:00 · **Stéphane Rivaud**
- This morning: forecasting (éCO2mix). Here: **anomalies**
- Then ~20 min: pretrained models (separate demo)

Supervised, or not

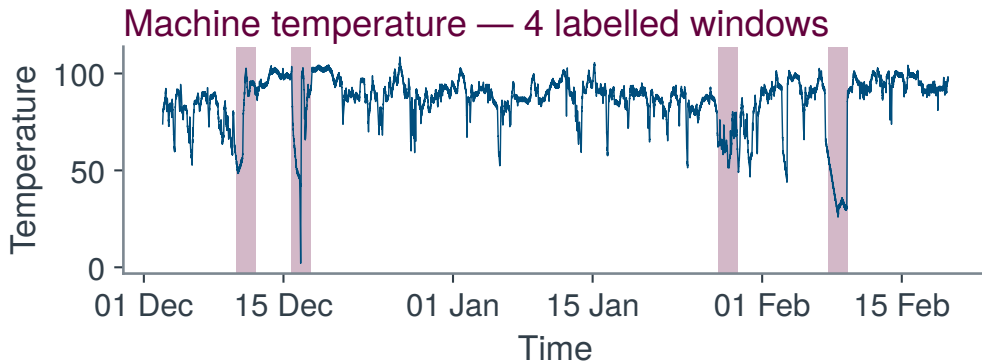
- **Supervised**: incident examples at train — here, 4 events
- **Unsupervised**: a score without a label (rolling z-score)
- **Semi-supervised**: the autoencoder learns the *normal*, not the failure
- NAB windows are used to **evaluate**, not to paste a target y

A machine temperature

NAB machine_temperature_system_failure · MIT · Numenta

- 5 min, from 2 December 2013 to 19 February 2014
- **22,683** points (12 duplicates on 7 January, first copy)
- **78.76** days = **11.25** weeks
- Median **89.4** · 2,268 points in a window = **10.0%**

Four windows, not 2,268 incidents



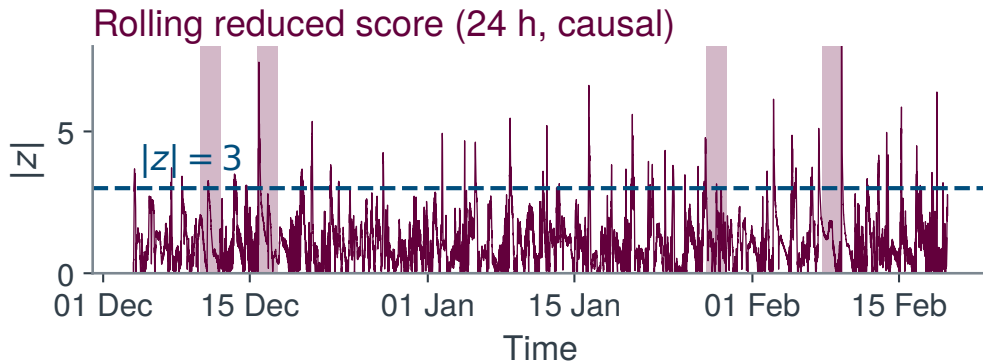
- 4 windows of **47 h** (567 points) · W2 falls to **2.08**
- W1 and W3 are lukewarm (troughs ~48–51) · W4 at **25.9**

Counting a detection

Lab convention — **say it**; others exist.

- A window is a hit if ≥ 1 alarm falls inside
- Ten alarms in the same window = **one** hit, not ten
- Alarms within ≤ 1 h = **one** event (on-call, not the 5 min step)
- Without merge: z-score prec. **0.065** and **15.3** FA/week (not 4.71)

The rolling z-score



- Past 24 h, **causal** (point t does not enter its reference)
- Classic threshold $|z| \geq 3 \cdot 1,008$ raw alarms

What the z-score gets

$|z| \geq 3$, 1 h merge, **the whole** series (11.25 weeks).

	Windows	prec.	FA / week
Events	4 / 4	0.102	4.71
Point-wise	recall 0.052	0.117	—

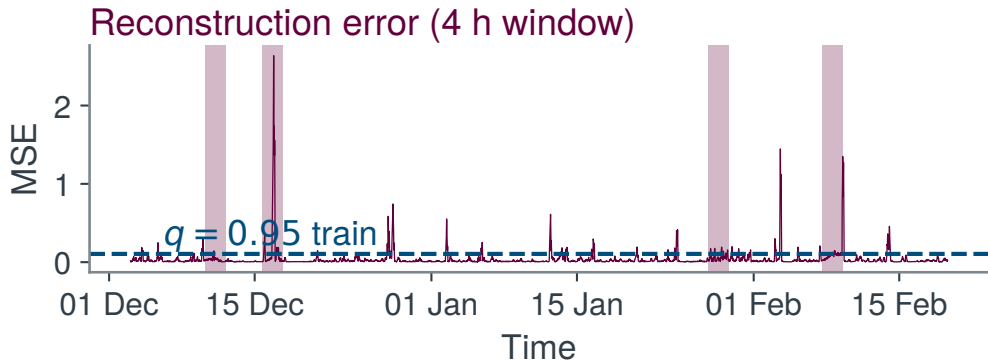
$|z| \geq 5$: recall **0.500** (W2, W4) · **0.80** FA/week · W1 and W3 **missed**.

The autoencoder reconstructs the normal

4 h windows (48 steps) $\rightarrow$ bottleneck 8 $\rightarrow$ reconstruction, MSE

- Train: **11,142** windows before **15 January**, outside labels
- Scaler fitted on this corpus ($\mu = 88.6$, $\sigma = 9.42$)
- 12 epochs, Adam, CPU · train loss = val = **0.030**
- This is not a classifier: no incident y

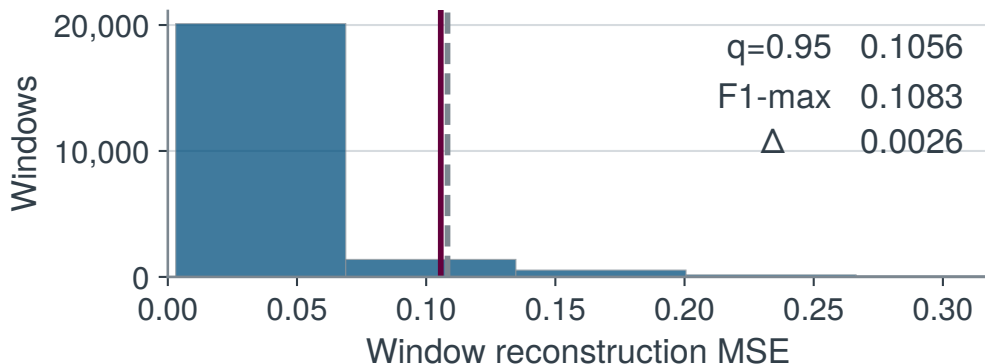
Reconstruction error



- Score = window MSE, aligned on **now**
- Threshold: **0.95** quantile of the **train** error (not the labels)

Quantile or F1

Train $q = 0.95$: no labels. F1-max **0.522** ($q=0.936$) — we **do not deploy** it.



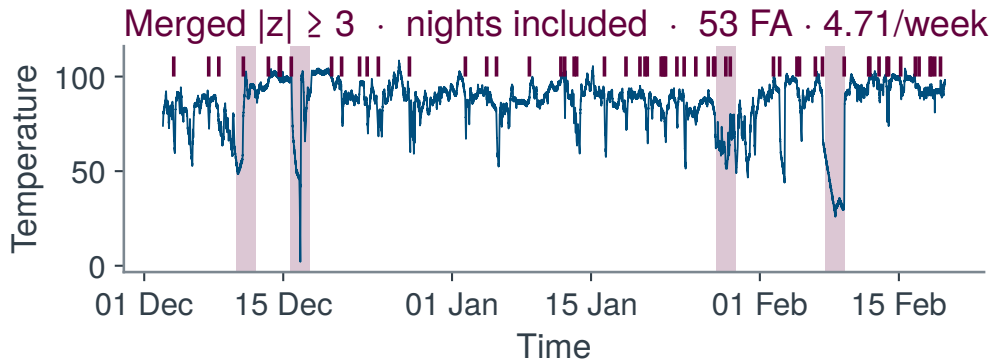
Z-score versus autoencoder

Same series, same window / event convention.

Policy	recall	prec.	FA / week
$ z \geq 3$	1.000	0.102	4.71
$ z \geq 5$	0.500	0.182	0.80
AE, $q = 0.95$	1.000	0.333	3.02
AE, F1 max	1.000	0.353	2.93

Same 4/4. The AE is a little **cleaner**, not another job.

Five false alarms per week



$|z| \geq 3$: **4.71** FA/week · AE **3.02** · $|z| \geq 5$: 0.80. No merge: 15, **unusable**.

What we would put in production

- **Deploy** the z-score $|z| \geq 3$, 1 h merge, on *this* series
- The AE: **shadow** (score logged, no page) — too thin a gain
- If 5 pages/week are too many: raise the threshold **and** a visual review
- To collect: true / false of **each** operator page

Not an API contest. A pager the shop floor can hold.

Instructions

- Default **autonomous** (guided if the room needs hints)
- File → Save a copy in Drive
- CPU
- Allowed everywhere · nothing to declare. Whatever you hand in, you wrote it — and you can explain every line.
- “Run all” without exercises: **NameError**, that is intended
- Groups of 3: the same until the oral exam of 5 October

Images: not a table of pixels

A convolution reads **neighbours**; flatten to an MLP and that structure is gone.

Road signs · Lyon



Yield sign



Car · Peugeot 207



Lorry



Bus · Paris



Train



Text: a sentence is not a row

- A sequence has an **order**; a bag-of-words throws it away
- Tokenise, contextualise: that is the job of **NLP** (M2)
- Here: do not treat an on-call ticket as 20 Excel features
- Same protocol: naive baseline before the fashionable model

A model already trained

Someone already paid for the pretraining (images, text, ...).

- We **freeze** (often) the trunk, we learn a **small head**
- Fewer labelled data at your place; less compute
- This is not “magic”: the domain must **look like** the source
- Demo next: the idea, not a benchmark to copy

What this changes for a project

A pretrained backbone **moves** the budget.

- Fewer weeks labelling; more weeks **sticking to the job**
- The cost: licence, drift vs ImageNet / the web, recaling
- A new MLP on 200 shop-floor photos **loses** this match
- Decision: reuse, adapt, or (rarely) train from scratch

Limits, and responsible AI

- **Data:** 4 events, one machine, one winter — not a fleet
- **Compute:** CPU today; an image backbone is another budget
- **Drift:** January's "normal" is not August's
- **Calibration:** a score is not a probability of incident

What follows is not a moral slide: UE **Responsible AI**, M2 S2.

Attributions and licences

Material and session: Stéphane Rivaud.

- [NAB](#) · Numenta Inc. · [MIT](#)
- [PyTorch tutorials](#) · BSD-3-Clause
- Photographs on “Images: not a table of pixels” · [CC0 1.0](#) · [images/ATTRIBUTION.md](#)

Series: `machine_temperature_system_failure` (`realKnownCause`).